

Lecture 9 (10/5)

Bias-variance trade off, confidence intervals and least squares

Bias and consistency

For any estimator, we define the mean squared error

$$\text{MSE}_{\hat{\theta}} = E[(\hat{\theta} - \theta)^2]$$

Bias and consistency

For any estimator, we define the mean squared error

$$\text{MSE}_{\hat{\theta}} = E[(\hat{\theta} - \theta)^2]$$

and we have the Bias-variance decomposition

$$\text{MSE}_{\hat{\theta}} = \text{se}(\hat{\theta})^2 + (E[\hat{\theta}] - \theta)^2.$$

Bias and consistency

For any estimator, we define the mean squared error

$$\text{MSE}_{\hat{\theta}} = E[(\hat{\theta} - \theta)^2]$$

and we have the Bias-variance decomposition

$$\text{MSE}_{\hat{\theta}} = \text{se}(\hat{\theta})^2 + (E[\hat{\theta}] - \theta)^2.$$

The bias is defined as $\text{Bias} = E[\hat{\theta}] - \theta$ and $\hat{\theta}$ is unbiased if the bias is zero.

Bias and consistency

For any estimator, we define the mean squared error

$$\text{MSE}_{\hat{\theta}} = E[(\hat{\theta} - \theta)^2]$$

and we have the Bias-variance decomposition

$$\text{MSE}_{\hat{\theta}} = \text{se}(\hat{\theta})^2 + (E[\hat{\theta}] - \theta)^2.$$

The bias is defined as $\text{Bias} = E[\hat{\theta}] - \theta$ and $\hat{\theta}$ is unbiased if the bias is zero. $\hat{\theta}$ is consistent if

$$E[\hat{\theta}] \rightarrow \theta \quad \text{and} \quad \text{se}(\hat{\theta}) \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Bias and consistency

For any estimator, we define the mean squared error

$$\text{MSE}_{\hat{\theta}} = E[(\hat{\theta} - \theta)^2]$$

and we have the Bias-variance decomposition

$$\text{MSE}_{\hat{\theta}} = \text{se}(\hat{\theta})^2 + (E[\hat{\theta}] - \theta)^2.$$

The bias is defined as $\text{Bias} = E[\hat{\theta}] - \theta$ and $\hat{\theta}$ is unbiased if the bias is zero. $\hat{\theta}$ is consistent if

$$E[\hat{\theta}] \rightarrow \theta \quad \text{and} \quad \text{se}(\hat{\theta}) \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Example

Last class we looked at the sample mean $\bar{Y}$ as an estimator both for Bernoulli and Normal. Is this unbiased and consistent?

Example

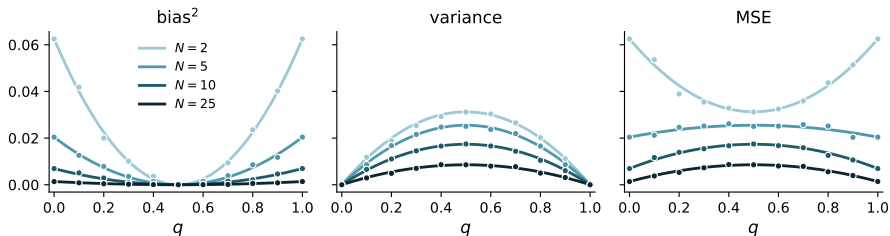
Let $Y_1, \dots, Y_N \sim \text{Bernoulli}(q)$. The natural estimator of q is $\hat{q} = \bar{Y}$. Thus, if every $Y_i = 0$, then $\hat{q} = 0$. This is problematic: If we flip the coin twice, this would lead us to conclude the coin can never land on heads. To correct for this, we pretend we have already seen two flips, one heads and one tails.

Laplace's rule of succession is the estimator

$$\hat{q}_L = \frac{S + 1}{N + 2}, \quad S = \sum_i Y_i.$$

Find the bias, variance and MSE of $\hat{q}_L$.

$$\text{bias} = \frac{1 - 2q}{N + 2}, \quad \text{se}(\hat{q}_L)^2 = \frac{Nq(1 - q)}{(N + 2)^2}$$



Curves are formulas above. Points are estimates from 1000 simulated replicates (see code on next slide).

```

N = 10
R = 1000 # number of replicates
qs = np.linspace(0, 1, 11)
for i in range(len(qs)):
    q = qs[i]
    q_hat = np.zeros(R)
    for r in range(R):
        S = np.random.binomial(N, q)
        # This also works:
        # ys = np.random.choice([0, 1], N, p=[1-q, q])
        # S = np.sum(ys)
        q_hat[r] = (S + 1) / (N + 2)
    bias2[i] = (np.mean(q_hat) - q)**2
    var[i] = np.var(q_hat)
    mse[i] = np.mean((q_hat - q)**2)

```

Each pass through the inner loop is one replicate of the data, so `q_hat` holds R draws from the sample distribution of $\hat{q}_L$. We repeat this for $N = 2, 5, 10, 25$.

Example

Suppose $Y_i \sim \text{Normal}(\mu, \sigma^2)$ and μ is known to us

$$\hat{\sigma}_\mu^2 = \frac{1}{N} \sum_i (Y_i - \mu)^2$$

Is this an unbiased estimator?

Example

Suppose $Y_i \sim \text{Normal}(\mu, \sigma^2)$ and μ is known to us

$$\hat{\sigma}_\mu^2 = \frac{1}{N} \sum_i (Y_i - \mu)^2$$

Is this an unbiased estimator?

Example

Suppose $Y_i \sim \text{Normal}(\mu, \sigma^2)$ and μ is NOT known to us

$$\hat{\sigma}_0^2 = \frac{1}{N} \sum_i (Y_i - \bar{Y})^2$$

Is this an unbiased estimator?